

WEEK 2 · SESSION 4

Encoders

Counting wheel turns to measure distance

Stop guessing how far the robot went. Measure it.

Time is only a proxy for distance

CARPET · 2 SECONDS

18 in

Higher friction slows the robot.

TILE · 2 SECONDS

26 in

Lower friction changes the result.

Same code + same time does not guarantee the same distance.

What is an encoder?

PATTERN

A striped or slotted disc turns with the motor shaft.

SENSOR

Each passing edge produces one count—a **tick**.

TWO SIDES

The XRP tracks left and right wheels independently.



Ticks → rotations

$$\text{rotations} = \frac{\text{ticks}}{\text{ticks per rotation}}$$

1440 TICKS

1 turn

2880 TICKS

2 turns

720 TICKS

½ turn

Use the exact XRP specification from your lab reference.

Rotations → distance

$$C = \pi d$$
$$x = \frac{N}{1440} C$$

WHEEL CIRCUMFERENCE

For $d \approx 2.4$ in, $C \approx 7.5$ in.

DISTANCE TRAVELED

N is the measured tick count.

One wheel rotation lays one circumference onto the floor.

Why encoders beat timing

TIMING-BASED

- Assumes a speed
- Changes with battery and floor
- Cannot verify wheel motion

ENCODER-BASED

- Measures actual wheel rotation
- Stops at a target count
- Repeatable across changing speed

Encoders measure the wheel—not the robot sliding over the floor.

Today's lab: compare three ways to move

1

EFFORT + TIMER

```
drivetrain.set_effort()
```

Run a fixed effort for a fixed time.

2

SPEED + TIMER

```
drivetrain.set_speed()
```

Use encoder-based speed control for the same timed move.

3

DISTANCE + TURNS

```
drivetrain.straight()
```

```
drivetrain.turn()
```

Command the movement goal directly.

Compare the accuracy and repeatability of each approach.

Count. Convert. Calibrate.

Encoder ticks turn hidden wheel motion into a number your program can use.

Your challenge: stop at 2 feet without using a timer.